

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with @BENCH adt_fields; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means SoA is faster; **bold** marks >1.1×.

Program	ADT	SoA	Fold passes			Map passes		
	fields	bufs	AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
OctTree	16	9	0.3740	0.3686	1.01×	0.3144	0.3582	0.88×
Compiler	9	8	0.1434	0.0812	1.77×	0.1616	0.1810	0.89×
DBQuery	15	13	0.2173	0.0917	2.37×	0.2081	0.1743	1.19×
DecisionTree	7	6	3.926	3.048	1.29×	—	—	—
DomTree	15	14	0.1436	0.0611	2.35×	0.0599	0.0827	0.72×
KDTree	17	16	0.1949	0.1610	1.21×	—	—	—
LinearListReduction	11	11	0.0320	0.0115	2.77×	—	—	—
List	2	2	0.1314	0.1111	1.18×	0.2548	0.3396	0.75×
MonoTree	3	2	0.0181	0.0224	0.81×	0.0465	0.0814	0.57×
ObjectGraph	5	4	0.0953	0.0918	1.04×	0.1476	0.1643	0.90×
PiecewiseFunctions	8	7	0.1354	0.0962	1.41×	0.2526	0.2744	0.92×
TernaryTree	5	3	0.0284	0.0257	1.10×	0.0870	0.0915	0.95×
Trie	10	9	0.0703	0.0454	1.55×	0.1532	0.1670	0.92×
ColorOctree	25	18	0.0890	0.0800	1.11×	—	—	—

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. >1× means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
OctTree	0.99×	0.75×	0.81×	0.46×	0.22×	0.63×	3.80×
Compiler	1.62×	0.52×	2.61×	0.84×	9.85×	0.33×	7.35×
DBQuery	1.00×	0.47×	1.38×	0.57×	8.17×	0.14×	5.26×
DecisionTree	1.28×	0.59×	17.83×	3.03×	0.93×	3.18×	35.92×
DomTree	1.54×	0.57×	1.88×	1.08×	20.70×	0.24×	9.96×
KDTree	1.40×	0.77×	1.33×	0.84×	0.28×	0.78×	4.45×
LinearListReduction	9.74×	0.90×	33.06×	2.64×	37.58×	3.92×	14.41×
List	0.82×	0.80×	3.37×	0.48×	0.20×	0.93×	1.48×
MonoTree	0.98×	0.85×	0.40×	0.82×	0.27×	1.07×	1.24×
ObjectGraph	0.98×	0.75×	6.07×	0.90×	0.54×	0.64×	2.75×
PiecewiseFunctions	1.07×	0.60×	2.31×	0.91×	1.83×	0.39×	4.46×
TernaryTree	1.15×	0.79×	0.32×	0.54×	0.18×	1.61×	1.32×
Trie	1.20×	0.56×	2.04×	0.84×	2.13×	0.18×	6.09×
Geomean	1.35×	0.67×	2.31×	0.89×	1.39×	0.66×	4.74×

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
OctTree	2.6e+09 /2.6e+09	8.2e+09 /1.1e+10	1.6e+07 /2.0e+07	3.7e+05 /8.1e+05	9.6e+04 /4.4e+05	1.7e+04 /2.7e+04	6.6e+07/ 1.7e+07
Compiler	9.5e+08/ 5.8e+08	9.9e+08 /1.9e+09	1.3e+07/ 5.1e+06	2.8e+05 /3.3e+05	1.3e+06/ 1.4e+05	1.1e+04 /3.3e+04	5.1e+07/ 6.9e+06
DBQuery	9.0e+08/ 8.9e+08	1.1e+09 /2.2e+09	4.6e+06/ 3.4e+06	2.2e+05 /3.8e+05	8.4e+05/ 1.0e+05	5.7e+03 /4.2e+04	2.0e+07/ 3.7e+06
DecisionTree	1.9e+10/ 1.5e+10	4.0e+10 /6.8e+10	5.3e+08/ 3.0e+07	1.4e+05/ 4.5e+04	2.0e+06 /2.1e+06	1.4e+05/ 4.3e+04	1.1e+09/ 3.0e+07
DomTree	8.5e+08/ 5.5e+08	1.2e+09 /2.2e+09	1.2e+07/ 6.4e+06	1.3e+05/ 1.2e+05	1.1e+06/ 5.2e+04	7.9e+03 /3.3e+04	5.7e+07/ 5.7e+06
KDTree	9.9e+08/ 7.0e+08	3.1e+09 /4.0e+09	1.4e+07/ 1.0e+07	1.9e+03 /2.2e+03	2.7e+04 /9.6e+04	1.5e+03 /2.0e+03	4.4e+07/ 9.9e+06
LinearListReduction	1.6e+08/ 1.7e+07	9.0e+07 /1.0e+08	1.5e+06/ 4.6e+04	9.4e+02/ 3.6e+02	2.7e+05/ 7.3e+03	8.2e+02/ 2.1e+02	1.2e+07/ 8.5e+05
List	1.3e+09 /1.5e+09	4.5e+09 /5.6e+09	9.6e+06/ 2.8e+06	2.6e+05 /5.5e+05	4.0e+04 /2.0e+05	9.7e+03 /1.0e+04	4.9e+07/ 3.3e+07
MonoTree	2.6e+08 /2.6e+08	9.1e+08 /1.1e+09	3.2e+05 /8.0e+05	3.9e+04 /4.8e+04	5.4e+03 /2.0e+04	3.1e+03/ 2.9e+03	2.3e+06/ 1.8e+06
ObjectGraph	8.6e+08 /8.7e+08	3.0e+09 /4.0e+09	1.0e+07/ 1.7e+06	2.2e+05 /2.4e+05	3.2e+04 /5.9e+04	1.1e+04 /1.6e+04	2.3e+07/ 8.5e+06
PiecewiseFunctions	1.2e+09/ 1.2e+09	2.8e+09 /4.7e+09	1.4e+07/ 6.2e+06	4.2e+05 /4.6e+05	2.0e+05/ 1.1e+05	2.1e+04 /5.4e+04	4.5e+07/ 1.0e+07
TernaryTree	4.1e+08/ 3.6e+08	1.2e+09 /1.5e+09	4.7e+05 /1.5e+06	6.6e+04 /1.2e+05	3.9e+03 /2.2e+04	3.6e+03/ 2.2e+03	3.3e+06/ 2.5e+06
Trie	6.9e+08/ 5.7e+08	1.2e+09 /2.2e+09	8.5e+06/ 4.2e+06	1.9e+05 /2.2e+05	1.6e+05/ 7.3e+04	7.7e+03 /4.3e+04	2.6e+07/ 4.3e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.3050	1.116	0.2622	<i>OOM</i>
DBQuery	0.4254	0.3423	0.2660	1.151
DecisionTree	3.926	5.696	3.048	25.39
DomTree	0.2035	0.3554	0.1438	0.5289
KDTree	0.1949	0.2523	0.1610	0.7329
LinearListReduction	0.0320	0.1378	0.0115	<i>OOM</i>
List	0.3863	<i>OOM</i>	0.4507	<i>OOM</i>
MonoTree	0.0646	0.1993	0.1038	0.2809
ObjectGraph	0.2429	0.5074	0.2561	0.7146
OctTree_sumMass	0.0352	0.0453	0.0296	<i>OOM</i>
OctTree_sumEnergy	0.0378	0.0552	0.0351	<i>OOM</i>
OctTree_countActive	0.0355	0.0505	0.0337	<i>OOM</i>
OctTree_countParticles	0.0373	0.0552	0.0308	<i>OOM</i>
OctTree_barnesHutPotential	0.0427	0.0606	0.0450	<i>OOM</i>
OctTree_fmmPotential	0.0964	0.0993	0.1145	<i>OOM</i>
OctTree_scaleEnergy	0.1548	0.1948	0.1794	<i>OOM</i>
OctTree_clearFlags	0.1595	0.1527	0.1788	<i>OOM</i>
PiecewiseFunctions	0.3879	0.5797	0.3706	2.375
TernaryTree	0.1154	0.1731	0.1171	0.6474
Trie	0.2235	0.3139	0.2125	1.002
ColorOctree	0.0890	0.1196	0.0800	<i>OOM</i>

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.16 $\times$	3.66 $\times$	4.26 $\times$	–
DBQuery	1.60 $\times$	0.80 $\times$	1.29 $\times$	0.30 $\times$
DecisionTree	1.29 $\times$	1.45 $\times$	1.87 $\times$	0.22 $\times$
DomTree	1.41 $\times$	1.75 $\times$	2.47 $\times$	0.67 $\times$
KDTree	1.21 $\times$	1.29 $\times$	1.57 $\times$	0.34 $\times$
LinearListReduction	2.77 $\times$	4.31 $\times$	11.94 $\times$	–
List	0.86 $\times$	–	–	–
MonoTree	0.62 $\times$	3.08 $\times$	1.92 $\times$	0.71 $\times$
ObjectGraph	0.95 $\times$	2.09 $\times$	1.98 $\times$	0.71 $\times$
OctTree_sumMass	1.19 $\times$	1.29 $\times$	1.53 $\times$	–
OctTree_sumEnergy	1.08 $\times$	1.46 $\times$	1.57 $\times$	–
OctTree_countActive	1.05 $\times$	1.42 $\times$	1.50 $\times$	–
OctTree_countParticles	1.21 $\times$	1.48 $\times$	1.79 $\times$	–
OctTree_barnesHutPotential	0.95 $\times$	1.42 $\times$	1.35 $\times$	–
OctTree_fmmPotential	0.84 $\times$	1.03 $\times$	0.87 $\times$	–
OctTree_scaleEnergy	0.86 $\times$	1.26 $\times$	1.09 $\times$	–
OctTree_clearFlags	0.89 $\times$	0.96 $\times$	0.85 $\times$	–
PiecewiseFunctions	1.05 $\times$	1.49 $\times$	1.56 $\times$	0.24 $\times$
TernaryTree	0.99 $\times$	1.50 $\times$	1.48 $\times$	0.27 $\times$
Trie	1.05 $\times$	1.40 $\times$	1.48 $\times$	0.31 $\times$
ColorOctree	1.11 $\times$	1.34 $\times$	1.50 $\times$	–

Table 6: Per-pass performance for **OctTree**, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
barnesHutPotential	F	11/16	31%	0.0427	0.0606	0.0450	–	0.95 $\times$	1.42 $\times$	1.35 $\times$	–
clearFlags	M	15/16	6%	0.1595	0.1527	0.1788	–	0.89 $\times$	0.96 $\times$	0.85 $\times$	–
countActive	F	10/16	38%	0.0355	0.0505	0.0337	–	1.05 $\times$	1.42 $\times$	1.50 $\times$	–
countParticles	F	8/16	50%	0.0373	0.0552	0.0308	–	1.21 $\times$	1.48 $\times$	1.79 $\times$	–
fmmPotential	F	12/16	25%	0.0964	0.0993	0.1145	–	0.84 $\times$	1.03 $\times$	0.87 $\times$	–
scaleEnergy	M	16/16	0%	0.1548	0.1948	0.1794	–	0.86 $\times$	1.26 $\times$	1.09 $\times$	–
sumEnergy	F	12/16	25%	0.0378	0.0552	0.0351	–	1.08 $\times$	1.46 $\times$	1.57 $\times$	–
sumMass	F	10/16	38%	0.0352	0.0453	0.0296	–	1.19 $\times$	1.29 $\times$	1.53 $\times$	–
paletteEntriesQuantized	F	13/25	48%	0.0428	0.0560	0.0370	–	1.15 $\times$	1.31 $\times$	1.51 $\times$	–
quantizationErrorProxy	F	10/25	60%	0.0462	0.0636	0.0429	–	1.08 $\times$	1.38 $\times$	1.48 $\times$	–
Total				0.6884	0.8331	0.7268	–	0.95 $\times$	1.21 $\times$	1.15 $\times$	–
Geomean								1.02 $\times$	1.29 $\times$	1.32 $\times$	–

Table 7: Per-pass PAPI counters for **OctTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
barnesHutPotential	F	2.2e+08 /2.2e+08	8.2e+08 /1.0e+09	7.4e+04 /5.7e+06	5.2e+02 /5.9e+02	2.8e+03 /1.1e+05	4.0e+02 /4.6e+02	6.2e+06/ 1.7e+06
clearFlags	M	3.7e+08 /4.5e+08	8.0e+08 /1.7e+09	1.3e+06 /2.6e+06	2.1e+05 /4.2e+05	9.9e+03 /4.6e+04	6.1e+03 /1.3e+04	5.6e+06/ 2.5e+06
countActive	F	1.8e+08/ 1.6e+08	5.8e+08/ 5.8e+08	9.9e+05/ 9.1e+05	7.5e+02/ 5.0e+02	4.1e+03 /2.3e+04	5.9e+02/ 3.1e+02	6.2e+06/ 3.0e+05
countParticles	F	1.9e+08/ 1.5e+08	5.7e+08/ 5.7e+08	9.2e+05/ 2.0e+05	6.6e+02/ 4.5e+02	3.8e+03 /8.2e+03	5.4e+02/ 2.0e+02	6.0e+06/ 1.4e+05
fmmPotential	F	4.9e+08/ 4.9e+08	1.6e+09 /1.8e+09	2.5e+04 /1.3e+06	5.6e+02 /1.4e+03	1.8e+03 /3.6e+04	4.3e+02 /1.2e+03	4.5e+06/ 2.2e+06
scaleEnergy	M	3.5e+08 /4.4e+08	9.1e+08 /1.9e+09	1.4e+06 /2.6e+06	1.6e+05 /3.9e+05	8.6e+03 /4.0e+04	7.2e+03 /1.1e+04	5.8e+06/ 2.5e+06
sumEnergy	F	1.8e+08/ 1.7e+08	7.7e+08 /8.9e+08	3.1e+05 /3.1e+06	5.8e+02/ 4.5e+02	2.1e+03 /6.7e+04	4.5e+02/ 2.9e+02	6.6e+06/ 2.4e+06
sumMass	F	1.7e+08/ 1.4e+08	5.7e+08 /6.3e+08	1.6e+06/ 2.1e+05	5.5e+02/ 4.0e+02	3.5e+04/ 1.0e+04	4.0e+02/ 2.6e+02	6.2e+06/ 1.3e+06
paletteEntriesQuantized	F	2.1e+08/ 1.8e+08	6.7e+08 /7.6e+08	5.2e+06/ 2.3e+06	5.0e+02 /5.6e+02	1.2e+04 /6.6e+04	4.1e+02 /4.2e+02	9.3e+06/ 1.3e+06
quantizationErrorProxy	F	2.3e+08/ 2.2e+08	9.7e+08 /1.2e+09	4.3e+06/ 9.3e+05	1.6e+02 /3.6e+02	1.6e+04 /3.4e+04	1.4e+02 /3.4e+02	9.2e+06/ 2.9e+06
Total		2.6e+09/2.6e+09	8.2e+09/1.1e+10	1.6e+07/2.0e+07	3.7e+05/8.1e+05	9.6e+04/4.4e+05	1.7e+04/2.7e+04	6.6e+07/1.7e+07

Table 8: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0125	0.0802	0.0037	<i>OOM</i>	3.35 $\times$	6.41 $\times$	21.51 $\times$	–
branchStatsPass	F	2/9	78%	0.0122	0.0811	0.0067	<i>OOM</i>	1.83 $\times$	6.66 $\times$	12.17 $\times$	–
castInstCountPass	F	3/9	67%	0.0186	0.0806	0.0032	<i>OOM</i>	5.74 $\times$	4.33 $\times$	24.83 $\times$	–
has cycle	F	4/9	56%	0.0352	0.0935	0.0400	<i>OOM</i>	0.88 $\times$	2.65 $\times$	2.34 $\times$	–
instCountPass	F	2/9	78%	0.0129	0.0798	0.0044	<i>OOM</i>	2.97 $\times$	6.17 $\times$	18.33 $\times$	–
latencyModelPass	F	3/9	67%	0.0124	0.0842	0.0047	<i>OOM</i>	2.61 $\times$	6.81 $\times$	17.79 $\times$	–
memoryOpStatsPass	F	3/9	67%	0.0123	0.0804	0.0090	<i>OOM</i>	1.36 $\times$	6.56 $\times$	8.94 $\times$	–
stripSideEffectsPass	M	7/9	22%	0.0809	0.1820	0.0914	<i>OOM</i>	0.89 $\times$	2.25 $\times$	1.99 $\times$	–
targetReturnPass	M	9/9	0%	0.0807	0.2013	0.0895	<i>OOM</i>	0.90 $\times$	2.50 $\times$	2.25 $\times$	–
throughputModelPass	F	3/9	67%	0.0124	0.0807	0.0031	<i>OOM</i>	4.06 $\times$	6.49 $\times$	26.32 $\times$	–
verifyIR	F	9/9	0%	0.0149	0.0716	0.0065	<i>OOM</i>	2.30 $\times$	4.82 $\times$	11.10 $\times$	–
Total				0.3050	1.116	0.2622	–	1.16 $\times$	3.66 $\times$	4.26 $\times$	–
Geomean								2.03 $\times$	4.68 $\times$	9.52 $\times$	–

Table 9: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	6.3e+07/ 1.0e+07	5.1e+07/ 4.5e+07	1.2e+06/ 1.0e+05	3.0e+02/ 8.4e+01	2.2e+05/ 2.6e+03	2.6e+02/ 7.2e+01	4.7e+06/ 3.5e+04
branchStatsPass	F	6.2e+07/ 1.5e+07	7.9e+07 /8.6e+07	8.1e+05/ 2.0e+04	2.2e+02/ 1.2e+02	5.4e+04/ 3.3e+03	1.8e+02/ 1.1e+02	4.7e+06/ 4.9e+05
castInstCountPass	F	9.4e+07/ 6.7e+06	5.1e+07/ 3.9e+07	1.3e+04 /3.0e+04	1.7e+02/ 1.4e+02	8.2e+02 /1.1e+03	1.5e+02/ 1.1e+02	4.7e+06/ 3.1e+04
has cycle	F	1.3e+08/ 8.7e+07	1.1e+08 /1.5e+08	2.4e+06/ 5.6e+05	8.3e+04/ 5.9e+04	1.2e+05/ 2.2e+04	2.1e+03/ 5.8e+02	5.0e+06/ 1.5e+06
instCountPass	F	5.6e+07/ 1.2e+07	5.6e+07 /5.7e+07	1.3e+06/ 1.0e+05	4.2e+02/ 1.2e+02	2.5e+05/ 2.7e+03	3.9e+02/ 1.0e+02	4.7e+06/ 3.8e+04
latencyModelPass	F	6.3e+07/ 1.1e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.3e+04	1.7e+02/ 1.0e+02	2.1e+05/ 4.0e+03	1.4e+02/ 7.5e+01	4.8e+06/ 5.6e+05
memoryOpStatsPass	F	6.2e+07/ 1.8e+07	8.5e+07 /9.1e+07	6.8e+05 /7.1e+05	2.0e+02/ 1.3e+02	3.4e+04/ 2.5e+04	1.8e+02/ 1.2e+02	4.7e+06/ 4.0e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	1.9e+06/ 1.3e+06	1.0e+05 /1.4e+05	5.6e+03 /2.0e+04	3.7e+03 /9.8e+03	4.3e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /1.9e+08	1.7e+08 /6.3e+08	1.5e+06/ 1.4e+06	8.9e+04 /1.3e+05	5.0e+03 /2.4e+04	2.8e+03 /2.1e+04	3.8e+06/ 1.5e+06
throughputModelPass	F	6.3e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 2.8e+04	2.8e+02/ 9.6e+01	2.1e+05/ 3.6e+03	2.5e+02/ 8.2e+01	4.8e+06/ 7.1e+05
verifyIR	F	5.0e+07/ 1.7e+07	7.4e+07 /8.6e+07	1.1e+06/ 7.8e+05	1.0e+03/ 7.9e+02	2.4e+05/ 2.8e+04	8.8e+02/ 6.1e+02	4.7e+06/ 4.5e+05
Total		9.5e+08/5.8e+08	9.9e+08/1.9e+09	1.3e+07/5.1e+06	2.8e+05/3.3e+05	1.3e+06/1.4e+05	1.1e+04/3.3e+04	5.1e+07/6.9e+06

Table 10: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.1e+08/−	1.6e+08/−	2.3e+06/−	6.1e+02/−	9.2e+04/−	5.8e+02/−	1.4e+07/−
branchStatsPass	F	4.1e+08/−	1.8e+08/−	2.1e+06/−	5.2e+02/−	4.5e+04/−	5.0e+02/−	1.4e+07/−
castInstCountPass	F	4.1e+08/−	1.8e+08/−	2.1e+06/−	5.4e+02/−	4.5e+04/−	5.0e+02/−	1.4e+07/−
has cycle	F	4.7e+08/−	2.1e+08/−	2.5e+06/−	3.7e+02/−	6.9e+04/−	3.6e+02/−	1.7e+07/−
instCountPass	F	4.1e+08/−	1.6e+08/−	2.3e+06/−	4.8e+02/−	9.7e+04/−	4.6e+02/−	1.4e+07/−
latencyModelPass	F	4.3e+08/−	1.7e+08/−	2.1e+06/−	5.3e+02/−	4.7e+04/−	4.9e+02/−	1.4e+07/−
memoryOpStatsPass	F	4.1e+08/−	1.9e+08/−	2.1e+06/−	3.5e+02/−	4.3e+04/−	3.2e+02/−	1.4e+07/−
stripSideEffectsPass	M	6.6e+08/−	3.3e+08/−	8.3e+05/−	9.9e+04/−	1.7e+04/−	1.1e+04/−	2.4e+07/−
targetReturnPass	M	6.5e+08/−	3.2e+08/−	1.1e+06/−	1.1e+05/−	1.9e+04/−	1.8e+04/−	2.2e+07/−
throughputModelPass	F	4.1e+08/−	1.7e+08/−	2.1e+06/−	1.0e+03/−	4.9e+04/−	9.8e+02/−	1.4e+07/−
verifyIR	F	3.6e+08/−	1.2e+08/−	2.5e+06/−	9.1e+02/−	2.1e+05/−	8.4e+02/−	1.1e+07/−
Total		5.0e+09/−	2.2e+09/−	2.2e+07/−	2.1e+05/−	7.3e+05/−	3.4e+04/−	1.7e+08/−

Table 11: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); ± shows standard error of the mean across −iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0729	0.0775	0.0846	0.3082	0.86×	1.06×	0.92×	0.25×
countJoins	F	3/15	80%	0.0193	0.0299	0.0117	0.0868	1.65×	1.55×	2.55×	0.34×
filterSelectivitySkew	F	5/15	67%	0.0753	0.0302	0.0132	0.0882	5.71×	0.40×	2.29×	0.34×
hashJoinPressure	F	5/15	67%	0.0209	0.0339	0.0133	0.0881	1.57×	1.62×	2.54×	0.38×
scaleCosts	M	15/15	0%	0.1352	0.0796	0.0897	0.3099	1.51×	0.59×	0.89×	0.26×
sumCost	F	6/15	60%	0.0418	0.0298	0.0154	0.0902	2.72×	0.71×	1.94×	0.33×
sumMemory	F	6/15	60%	0.0191	0.0297	0.0157	0.0901	1.22×	1.56×	1.89×	0.33×
sumRows	F	6/15	60%	0.0409	0.0317	0.0224	0.0894	1.83×	0.78×	1.42×	0.35×
Total				0.4254	0.3423	0.2660	1.151	1.60×	0.80×	1.29×	0.30×
Geomean								1.81×	0.93×	1.68×	0.32×

Table 12: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.7e+08 /2.2e+08	1.9e+08 /7.2e+08	2.9e+05 /1.5e+06	1.1e+05 /1.8e+05	3.1e+04/ 2.7e+04	1.3e+03 /1.5e+04	2.4e+06/ 9.4e+05
countJoins	F	9.6e+07/ 5.8e+07	7.4e+07/ 7.1e+07	6.1e+05/ 2.5e+04	4.1e+02/ 2.4e+02	1.4e+05/ 2.0e+03	3.8e+02/ 1.8e+02	2.7e+06/ 6.2e+04
filterSelectivitySkew	F	7.9e+07/ 6.5e+07	1.0e+08 /1.1e+08	5.7e+05/ 7.7e+04	3.2e+02/ 1.9e+02	1.2e+05/ 6.1e+03	3.0e+02/ 1.8e+02	2.0e+06/ 3.6e+05
hashJoinPressure	F	1.0e+08/ 6.6e+07	8.4e+07 /9.9e+07	1.3e+06/ 9.9e+04	7.5e+02/ 1.4e+02	2.2e+05/ 7.5e+03	7.2e+02/ 1.2e+02	2.8e+06/ 1.5e+05
scaleCosts	M	1.6e+08 /2.2e+08	2.2e+08 /7.5e+08	2.9e+05 /1.4e+06	1.0e+05 /1.9e+05	3.2e+04/ 3.0e+04	2.1e+03 /2.6e+04	2.3e+06/ 1.1e+06
sumCost	F	8.3e+07/ 7.6e+07	9.0e+07 /1.2e+08	5.9e+05/ 1.7e+05	4.6e+02/ 3.8e+02	1.2e+05/ 1.2e+04	3.6e+02/ 2.6e+02	2.3e+06/ 3.9e+05
sumMemory	F	9.5e+07/ 7.8e+07	9.0e+07 /1.2e+08	6.7e+05/ 7.9e+04	3.2e+02/ 1.8e+02	1.5e+05/ 1.1e+04	3.1e+02/ 1.7e+02	2.8e+06/ 3.9e+05
sumRows	F	1.2e+08/ 1.1e+08	2.1e+08 /2.3e+08	3.1e+05/ 1.0e+05	2.0e+02 /2.1e+02	2.4e+04/ 7.9e+03	1.7e+02 /1.9e+02	2.2e+06/ 3.3e+05
Total		9.0e+08/8.9e+08	1.1e+09/2.2e+09	4.6e+06/3.4e+06	2.2e+05/3.8e+05	8.4e+05/1.0e+05	5.7e+03/4.2e+04	2.0e+07/3.7e+06

Table 13: Per-pass PAPI counters for **DBQuery** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.0e+08 /1.3e+09	3.2e+08 /2.4e+09	6.4e+05 /2.0e+07	9.4e+04 /6.9e+05	1.5e+04 /5.5e+04	2.8e+03 /4.5e+04	2.1e+06/ 1.0e+06
countJoins	F	1.5e+08 /4.3e+08	1.6e+08 /7.7e+08	1.9e+06/ 3.2e+05	3.1e+02 /1.0e+03	1.2e+05/ 1.3e+04	2.9e+02/ 1.9e+02	2.5e+06/ 9.3e+04
filterSelectivitySkew	F	1.5e+08 /4.4e+08	2.0e+08 /7.9e+08	1.4e+06/ 8.4e+05	3.6e+02 /1.8e+03	6.6e+04/ 1.7e+04	3.2e+02 /6.3e+02	2.4e+06/ 2.7e+05
hashJoinPressure	F	1.7e+08 /4.4e+08	1.8e+08 /7.7e+08	1.6e+06/ 6.5e+05	2.1e+02 /1.2e+03	1.1e+05/ 1.6e+04	1.8e+02 /4.0e+02	2.3e+06/ 2.0e+05
scaleCosts	M	2.0e+08 /1.3e+09	3.5e+08 /2.4e+09	5.6e+05 /2.3e+07	7.9e+04 /3.1e+05	1.8e+04 /5.5e+04	3.5e+03 /5.9e+04	2.3e+06/ 1.2e+06
sumCost	F	1.5e+08 /4.5e+08	1.8e+08 /7.8e+08	1.7e+06/ 1.2e+06	6.9e+02 /1.7e+03	7.6e+04/ 2.3e+04	5.8e+02 /1.0e+03	2.4e+06/ 3.6e+05
sumMemory	F	1.5e+08 /4.5e+08	1.8e+08 /7.8e+08	1.5e+06/ 1.1e+06	2.8e+02 /1.4e+03	7.1e+04/ 2.1e+04	2.5e+02 /7.3e+02	2.4e+06/ 3.6e+05
sumRows	F	1.6e+08 /4.4e+08	2.3e+08 /8.3e+08	1.5e+06/ 1.2e+06	3.0e+02 /1.6e+03	5.9e+04/ 2.2e+04	2.6e+02 /9.4e+02	2.3e+06/ 3.7e+05
Total		1.3e+09/5.3e+09	1.8e+09/9.5e+09	1.1e+07/4.9e+07	1.7e+05/1.0e+06	5.3e+05/2.2e+05	8.1e+03/1.1e+05	1.9e+07/3.9e+06

Table 14: Per-pass performance for **DecisionTree**, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); ± shows standard error of the mean across −iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	3.374	4.316	2.699	21.40	1.25×	1.28×	1.60×	0.20×
classify Depth	F	4/7	43%	0.0465	0.0619	0.0389	0.2989	1.20×	1.33×	1.59×	0.21×
classify tree	F	5/7	29%	0.0464	0.0618	0.0390	0.3053	1.19×	1.33×	1.58×	0.20×
countFeatureUses	F	3/7	57%	0.0502	0.1420	0.0323	0.3789	1.56×	2.83×	4.40×	0.37×
countLeaves	F	2/7	71%	0.0477	0.1405	0.0249	0.3696	1.92×	2.95×	5.65×	0.38×
countNodes	F	2/7	71%	0.0480	0.1446	0.0263	0.3706	1.83×	3.01×	5.50×	0.39×
countSmallLeaves	F	3/7	57%	0.0529	0.1365	0.0323	0.3720	1.63×	2.58×	4.22×	0.37×
inferenceCost	F	2/7	71%	0.0508	0.1395	0.0291	0.3705	1.75×	2.75×	4.79×	0.38×
sumImpurity	F	3/7	57%	0.0502	0.1392	0.0326	0.3813	1.54×	2.77×	4.26×	0.37×
sumPathLengths	F	3/7	57%	0.0532	0.1392	0.0337	0.3875	1.58×	2.62×	4.14×	0.36×
sumSamples	F	3/7	57%	0.0536	0.1359	0.0320	0.3842	1.67×	2.54×	4.25×	0.35×
treeDepth	F	2/7	71%	0.0515	0.1394	0.0280	0.3701	1.84×	2.71×	4.98×	0.38×
Total				3.926	5.696	3.048	25.39	1.29×	1.45×	1.87×	0.22×
Geomean								1.56×	2.28×	3.56×	0.32×

Table 15: Per-pass PAPI counters for **DecisionTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.6e+10/ 1.3e+10	3.3e+10 /5.9e+10	4.6e+08/ 2.7e+07	1.2e+05/ 3.6e+04	1.6e+06 /2.0e+06	1.2e+05/ 3.4e+04	9.3e+08/ 1.8e+07
classify Depth	F	2.3e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.6e+06/ 4.1e+05	8.6e+02/ 5.2e+02	2.3e+04 /2.8e+04	8.3e+02/ 4.7e+02	1.3e+07/ 3.3e+05
classify tree	F	2.3e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.6e+06/ 4.1e+05	9.2e+02/ 5.3e+02	2.2e+04 /2.8e+04	9.0e+02/ 4.7e+02	1.3e+07/ 3.4e+05
countFeatureUses	F	2.5e+08/ 1.6e+08	7.5e+08 /7.8e+08	7.3e+06/ 3.0e+05	1.8e+03/ 7.2e+02	3.7e+04/ 1.3e+04	1.7e+03/ 6.9e+02	1.5e+07/ 2.1e+06
countLeaves	F	2.4e+08/ 1.3e+08	6.0e+08 /6.1e+08	7.2e+06/ 1.0e+05	9.7e+02/ 6.6e+02	4.5e+04/ 7.9e+03	9.3e+02/ 5.8e+02	1.4e+07/ 4.2e+05
countNodes	F	2.4e+08/ 1.3e+08	6.1e+08 /6.2e+08	7.0e+06/ 1.1e+05	1.3e+03 /1.6e+03	5.3e+04/ 8.0e+03	1.2e+03 /1.5e+03	1.4e+07/ 4.3e+05
countSmallLeaves	F	2.6e+08/ 1.6e+08	7.1e+08 /8.4e+08	5.9e+06/ 9.2e+04	2.1e+03/ 8.7e+02	3.4e+04/ 1.3e+04	2.1e+03/ 8.5e+02	1.5e+07/ 2.0e+06
inferenceCost	F	2.5e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.0e+06/ 1.2e+05	1.7e+03/ 1.2e+03	1.9e+04/ 8.1e+03	1.6e+03/ 1.1e+03	1.4e+07/ 4.1e+05
sumImpurity	F	2.5e+08/ 1.7e+08	7.6e+08 /7.8e+08	7.5e+06/ 1.0e+06	1.1e+03/ 6.8e+02	3.7e+04/ 1.3e+04	1.0e+03/ 6.6e+02	1.5e+07/ 2.1e+06
sumPathLengths	F	2.6e+08/ 1.7e+08	8.0e+08 /9.0e+08	7.5e+06/ 5.0e+05	1.6e+03/ 1.1e+03	3.7e+04/ 1.4e+04	1.6e+03/ 1.0e+03	1.5e+07/ 2.0e+06
sumSamples	F	2.6e+08/ 1.6e+08	6.4e+08 /7.7e+08	6.0e+06/ 2.4e+05	3.6e+03/ 6.2e+02	4.1e+04/ 1.4e+04	3.5e+03/ 6.0e+02	1.5e+07/ 2.0e+06
treeDepth	F	2.6e+08/ 1.4e+08	6.1e+08/ 6.1e+08	7.9e+06/ 1.1e+05	1.1e+03/ 8.9e+02	2.0e+04/ 8.0e+03	1.0e+03/ 8.2e+02	1.4e+07/ 4.2e+05
Total		1.9e+10/1.5e+10	4.0e+10/6.8e+10	5.3e+08/3.0e+07	1.4e+05/4.5e+04	2.0e+06/2.1e+06	1.4e+05/4.3e+04	1.1e+09/3.0e+07

Table 16: Per-pass PAPI counters for **DecisionTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	2.2e+10 /1.1e+11	5.1e+10 /1.9e+11	1.5e+07 /4.5e+07	2.5e+04 /1.4e+05	3.6e+05 /2.1e+06	2.5e+04/ 2.4e+04	8.6e+08/ 2.3e+07
classify Depth	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.6e+05 /6.6e+05	3.3e+02 /1.8e+03	4.8e+03 /3.2e+04	3.0e+02/ 3.0e+02	1.2e+07/ 3.3e+05
classify tree	F	3.1e+08 /1.6e+09	7.4e+08 /2.7e+09	2.5e+05 /6.7e+05	2.8e+02 /2.3e+03	4.8e+03 /3.2e+04	2.5e+02 /4.4e+02	1.2e+07/ 3.3e+05
countFeatureUses	F	7.2e+08 /1.9e+09	1.3e+09 /3.1e+09	2.0e+06 /3.2e+06	3.0e+02 /2.6e+03	1.1e+04 /7.0e+04	2.8e+02 /1.1e+03	9.4e+06/ 1.2e+06
countLeaves	F	7.1e+08 /1.9e+09	1.2e+09 /2.9e+09	4.3e+06/ 6.7e+05	1.0e+03 /3.0e+03	1.5e+04 /3.2e+04	9.8e+02/ 4.7e+02	9.2e+06/ 3.8e+05
countNodes	F	7.2e+08 /1.9e+09	1.2e+09 /3.0e+09	4.7e+06/ 6.8e+05	1.4e+03 /2.6e+03	2.6e+04 /3.2e+04	1.2e+03/ 6.0e+02	9.2e+06/ 3.8e+05
countSmallLeaves	F	6.9e+08 /1.9e+09	1.3e+09 /2.9e+09	1.1e+06 /3.4e+06	3.6e+02 /2.2e+03	7.1e+03 /7.4e+04	3.3e+02 /8.7e+02	9.3e+06/ 1.2e+06
inferenceCost	F	7.1e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.2e+05	6.4e+02 /1.6e+03	1.4e+04 /3.1e+04	6.2e+02/ 3.6e+02	9.2e+06/ 3.8e+05
sumImpurity	F	7.1e+08 /1.9e+09	1.3e+09 /3.0e+09	2.2e+06 /3.5e+06	2.6e+02 /4.4e+02	1.2e+04 /7.5e+04	2.3e+02 /3.7e+02	9.5e+06/ 1.2e+06
sumPathLengths	F	7.1e+08 /2.0e+09	1.3e+09 /3.1e+09	1.5e+06 /3.6e+06	5.7e+02 /4.7e+03	7.5e+03 /8.0e+04	5.5e+02 /1.9e+03	9.2e+06/ 1.2e+06
sumSamples	F	6.9e+08 /2.0e+09	1.2e+09 /3.0e+09	1.1e+06 /3.6e+06	9.1e+02 /2.4e+03	7.2e+03 /7.8e+04	8.8e+02 /1.0e+03	9.3e+06/ 1.2e+06
treeDepth	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.2e+05	4.3e+02 /1.7e+03	1.4e+04 /3.1e+04	3.9e+02/ 3.6e+02	9.2e+06/ 4.0e+05
Total		2.9e+10/1.3e+11	6.4e+10/2.2e+11	4.1e+07/6.6e+07	3.1e+04/1.6e+05	4.8e+05/2.7e+06	3.1e+04/3.2e+04	9.7e+08/3.1e+07

Table 17: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0373	0.0834	0.0202	0.0913	1.84 $\times$	2.24 $\times$	4.12 $\times$	0.91 $\times$
computeWidths	M	14/15	7%	0.0304	0.0305	0.0463	0.0881	0.66 $\times$	1.00 $\times$	0.66 $\times$	0.35 $\times$
count styled	F	3/15	80%	0.0359	0.0702	0.0117	0.0881	3.07 $\times$	1.96 $\times$	6.01 $\times$	0.80 $\times$
find max Bottom	F	5/15	67%	0.0354	0.0699	0.0184	0.0872	1.93 $\times$	1.97 $\times$	3.80 $\times$	0.80 $\times$
scaleLayout	M	15/15	0%	0.0295	0.0307	0.0364	0.0870	0.81 $\times$	1.04 $\times$	0.84 $\times$	0.35 $\times$
sumTextWidth	F	3/15	80%	0.0350	0.0708	0.0109	0.0873	3.22 $\times$	2.02 $\times$	6.53 $\times$	0.81 $\times$
Total				0.2035	0.3554	0.1438	0.5289	1.41 $\times$	1.75 $\times$	2.47 $\times$	0.67 $\times$
Geomean								1.63 $\times$	1.62 $\times$	2.64 $\times$	0.62 $\times$

Table 18: Per-pass PAPI counters for **DomTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.9e+08/ 9.9e+07	2.9e+08 /4.9e+08	2.6e+06/ 1.0e+06	9.8e+02/ 5.0e+02	1.6e+05/ 1.4e+04	8.6e+02/ 3.9e+02	1.4e+07/ 1.9e+06
computeWidths	M	6.6e+07 /1.5e+08	1.4e+08 /4.6e+08	1.2e+06 /1.6e+06	6.5e+04 /6.6e+04	4.3e+03 /9.5e+03	1.4e+03 /1.7e+04	1.3e+06/ 4.6e+05
count styled	F	1.8e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.1e+06/ 2.4e+04	1.7e+03/ 1.1e+02	2.9e+05/ 4.9e+03	1.6e+03/ 9.9e+01	1.4e+07/ 7.2e+05
find max Bottom	F	1.8e+08/ 9.3e+07	2.5e+08 /4.1e+08	2.5e+06/ 1.4e+06	4.0e+02/ 1.9e+02	1.6e+05/ 9.9e+03	3.8e+02/ 1.6e+02	1.4e+07/ 1.5e+06
scaleLayout	M	5.9e+07 /9.8e+07	8.8e+07 /2.9e+08	1.1e+06 /2.3e+06	5.8e+04/ 5.0e+04	2.4e+04/ 1.0e+04	2.5e+03 /1.5e+04	1.4e+06/ 4.4e+05
sumTextWidth	F	1.8e+08/ 5.5e+07	2.0e+08 /2.4e+08	2.5e+06/ 1.8e+04	1.2e+03/ 2.5e+02	4.4e+05/ 4.4e+03	1.2e+03/ 2.4e+02	1.4e+07/ 7.4e+05
Total		8.5e+08/5.5e+08	1.2e+09/2.2e+09	1.2e+07/6.4e+06	1.3e+05/1.2e+05	1.1e+06/5.2e+04	7.9e+03/3.3e+04	5.7e+07/5.7e+06

Table 19: Per-pass PAPI counters for **DomTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	4.2e+08 /4.5e+08	4.4e+08 /1.8e+09	1.3e+07/ 2.5e+06	5.8e+02 /1.6e+03	4.0e+05/ 4.0e+04	4.6e+02 /1.4e+03	9.2e+06/ 1.4e+06
computeWidths	M	6.8e+07 /3.6e+08	1.5e+08 /9.0e+08	1.3e+06 /3.4e+06	4.7e+04 /2.5e+05	4.0e+03 /1.5e+04	1.3e+03 /2.3e+04	1.2e+06/ 4.7e+05
count styled	F	3.6e+08 /4.5e+08	4.3e+08 /1.7e+09	1.0e+07/ 1.3e+06	2.2e+02 /4.5e+02	3.7e+04/ 3.7e+04	2.2e+02 /4.1e+02	1.0e+07/ 4.5e+05
find max Bottom	F	3.5e+08 /4.4e+08	4.6e+08 /1.8e+09	1.0e+07/ 1.5e+06	1.7e+02 /9.1e+02	1.5e+04 /2.6e+04	1.5e+02 /8.9e+02	1.0e+07/ 1.0e+06
scaleLayout	M	6.8e+07 /3.5e+08	1.1e+08 /7.6e+08	9.8e+05 /4.1e+06	5.4e+04 /8.4e+04	8.5e+03 /1.0e+04	1.2e+03 /2.3e+04	1.2e+06/ 4.6e+05
sumTextWidth	F	3.6e+08 /4.4e+08	3.9e+08 /1.7e+09	1.0e+07/ 1.3e+06	2.9e+02 /1.0e+03	8.9e+04/ 3.9e+04	2.4e+02 /8.1e+02	1.0e+07/ 4.4e+05
Total		1.6e+09/2.5e+09	2.0e+09/8.6e+09	4.6e+07/1.4e+07	1.0e+05/3.3e+05	5.6e+05/1.7e+05	3.5e+03/5.0e+04	4.2e+07/4.2e+06

Table 20: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	13/17	24%	0.0309	0.0455	0.0403	0.1199	0.77 $\times$	1.47 $\times$	1.13 $\times$	0.38 $\times$
countInRange tight_box	F	13/17	24%	0.0306	0.0403	0.0222	0.1266	1.37 $\times$	1.32 $\times$	1.81 $\times$	0.32 $\times$
photonMappingBenchmark	F	12/17	29%	0.0451	0.0479	0.0410	0.1336	1.10 $\times$	1.06 $\times$	1.17 $\times$	0.36 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0273	0.0389	0.0165	0.1142	1.66 $\times$	1.42 $\times$	2.36 $\times$	0.34 $\times$
sumMassInRange	F	14/17	18%	0.0298	0.0405	0.0203	0.1248	1.47 $\times$	1.36 $\times$	2.00 $\times$	0.32 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0313	0.0392	0.0207	0.1138	1.51 $\times$	1.25 $\times$	1.89 $\times$	0.34 $\times$
Total				0.1949	0.2523	0.1610	0.7329	1.21 $\times$	1.29 $\times$	1.57 $\times$	0.34 $\times$
Geomean								1.27 $\times$	1.31 $\times$	1.67 $\times$	0.34 $\times$

Table 21: Per-pass PAPI counters for **KDTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.5e+08/ 1.0e+08	4.7e+08 /6.3e+08	2.8e+06/ 1.7e+06	5.9e+02 /6.2e+02	8.7e+03 /1.8e+04	4.3e+02 /5.0e+02	7.7e+06/ 1.5e+06
countInRange tight_box	F	1.5e+08/ 9.9e+07	4.0e+08 /5.5e+08	2.0e+06 /2.1e+06	2.8e+02/ 1.8e+02	3.0e+03 /1.7e+04	2.4e+02/ 1.6e+02	7.7e+06/ 1.7e+06
photonMappingBenchmark	F	2.3e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.2e+06 /2.5e+06	3.8e+02 /5.9e+02	3.7e+03 /1.4e+04	3.3e+02 /5.8e+02	5.7e+06/ 1.5e+06
pointCloudNeighborhood	F	1.4e+08/ 8.3e+07	3.3e+08 /4.8e+08	2.6e+06/ 8.0e+05	1.9e+02/ 1.6e+02	5.2e+03 /1.5e+04	1.7e+02/ 1.3e+02	7.8e+06/ 1.7e+06
sumMassInRange	F	1.5e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.2e+06 /2.7e+06	2.0e+02 /2.1e+02	3.7e+03 /1.9e+04	1.6e+02 /2.0e+02	7.7e+06/ 2.1e+06
twoPointCorrelation bin_8_16	F	1.6e+08/ 1.0e+08	4.8e+08 /6.1e+08	2.2e+06/ 6.5e+05	2.2e+02 /4.3e+02	3.1e+03 /1.3e+04	2.0e+02 /4.2e+02	7.7e+06/ 1.4e+06
Total		9.9e+08/7.0e+08	3.1e+09/4.0e+09	1.4e+07/1.0e+07	1.9e+03/2.2e+03	2.7e+04/9.6e+04	1.5e+03/2.0e+03	4.4e+07/9.9e+06

Table 22: Per-pass PAPI counters for **KDTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.0e+08 /6.0e+08	4.9e+08 /1.3e+09	5.0e+06 /5.4e+06	5.4e+02 /2.5e+03	3.2e+04 /3.4e+04	3.4e+02 /2.3e+03	5.5e+06/ 1.5e+06
countInRange tight_box	F	2.0e+08 /6.3e+08	4.5e+08 /1.3e+09	5.1e+06 /5.8e+06	2.0e+02 /3.2e+03	5.7e+03 /2.4e+04	1.9e+02 /3.1e+03	5.7e+06/ 1.6e+06
photonMappingBenchmark	F	2.4e+08 /6.8e+08	1.1e+09 /2.0e+09	2.8e+06 /5.9e+06	4.7e+02 /2.1e+03	2.8e+03 /2.6e+04	4.6e+02 /2.1e+03	5.2e+06/ 1.5e+06
pointCloudNeighborhood	F	2.0e+08 /5.8e+08	4.1e+08 /1.3e+09	5.1e+06/ 4.7e+06	2.5e+02 /1.6e+03	1.3e+04 /2.0e+04	2.4e+02 /1.4e+03	5.9e+06/ 1.4e+06
sumMassInRange	F	2.0e+08 /6.3e+08	4.7e+08 /1.3e+09	5.1e+06 /6.3e+06	2.6e+02 /1.9e+03	5.2e+03 /2.8e+04	2.4e+02 /1.8e+03	5.7e+06/ 1.7e+06
twoPointCorrelation bin_8_16	F	2.0e+08 /5.8e+08	6.1e+08 /1.4e+09	4.8e+06/ 3.7e+06	3.0e+02 /2.0e+03	6.1e+03 /1.9e+04	2.8e+02 /2.0e+03	5.8e+06/ 1.3e+06
Total		1.2e+09/3.7e+09	3.5e+09/8.4e+09	2.8e+07/3.2e+07	2.0e+03/1.3e+04	6.5e+04/1.5e+05	1.8e+03/1.3e+04	3.4e+07/9.0e+06

Table 23: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0320	0.1378	0.0115	<i>OOM</i>	2.77 $\times$	4.31 $\times$	11.94 $\times$	–
Total				0.0320	0.1378	0.0115	–	2.77 $\times$	4.31 $\times$	11.94 $\times$	–
Geomean								2.77 $\times$	4.31 $\times$	11.94 $\times$	–

Table 24: Per-pass PAPI counters for **LinearListReduction** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.6e+08/ 1.7e+07	9.0e+07 /1.0e+08	1.5e+06/ 4.6e+04	9.4e+02/ 3.6e+02	2.7e+05/ 7.3e+03	8.2e+02/ 2.1e+02	1.2e+07/ 8.5e+05
Total		1.6e+08/1.7e+07	9.0e+07/1.0e+08	1.5e+06/4.6e+04	9.4e+02/3.6e+02	2.7e+05/7.3e+03	8.2e+02/2.1e+02	1.2e+07/8.5e+05

Table 25: Per-pass PAPI counters for **LinearListReduction** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	6.9e+08/–	2.5e+08/–	5.0e+06/–	1.1e+03/–	5.5e+05/–	8.9e+02/–	2.6e+07/–
Total		6.9e+08/–	2.5e+08/–	5.0e+06/–	1.1e+03/–	5.5e+05/–	8.9e+02/–	2.6e+07/–

Table 26: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.2548	<i>OOM</i>	0.3396	<i>OOM</i>	0.75 $\times$	–	–	–
length List	F	1/2	50%	0.0426	<i>OOM</i>	0.0216	<i>OOM</i>	1.97 $\times$	–	–	–
sumList	F	2/2	0%	0.0442	<i>OOM</i>	0.0451	<i>OOM</i>	0.98 $\times$	–	–	–
sumList tail recursive	F	2/2	0%	0.0446	<i>OOM</i>	0.0445	<i>OOM</i>	1.00 $\times$	–	–	–
Total				0.3863	–	0.4507	–	0.86 $\times$	–	–	–
Geomean								1.10 $\times$	–	–	–

Table 27: Per-pass PAPI counters for **List** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.7e+08	1.9e+09 /2.8e+09	8.4e+06/ 1.6e+06	2.6e+05 /5.5e+05	3.1e+04 /6.0e+04	9.2e+03 /1.0e+04	9.8e+06/ 7.7e+06
length List	F	2.2e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.3e+05/ 9.3e+04	1.9e+02/ 8.4e+01	2.9e+03 /1.4e+04	1.6e+02/ 8.0e+01	1.3e+07/ 1.2e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	6.2e+05/ 5.4e+05	1.6e+02/ 1.1e+02	3.1e+03 /6.9e+04	1.5e+02/ 1.0e+02	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.3e+08/ 2.3e+08	9.0e+08 /1.0e+09	4.3e+05 /5.7e+05	1.9e+02/ 1.7e+02	3.0e+03 /5.8e+04	1.6e+02 /1.6e+02	1.3e+07/ 1.2e+07
Total		1.3e+09/1.5e+09	4.5e+09/5.6e+09	9.6e+06/2.8e+06	2.6e+05/5.5e+05	4.0e+04/2.0e+05	9.7e+03/1.0e+04	4.9e+07/3.3e+07

Table 28: Per-pass PAPI counters for **List** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		–/–	–/–	–/–	–/–	–/–	–/–	–/–

Table 29: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0465	0.1186	0.0814	0.1193	0.57 $\times$	2.55 $\times$	1.46 $\times$	0.99 $\times$
sumTree	F	3/3	0%	0.0093	0.0401	0.0112	0.0807	0.83 $\times$	4.30 $\times$	3.57 $\times$	0.50 $\times$
sumTree TailRec	F	3/3	0%	0.0088	0.0405	0.0111	0.0809	0.79 $\times$	4.62 $\times$	3.64 $\times$	0.50 $\times$
Total				0.0646	0.1993	0.1038	0.2809	0.62 $\times$	3.08 $\times$	1.92 $\times$	0.71 $\times$
Geomean								0.72 $\times$	3.70 $\times$	2.66 $\times$	0.63 $\times$

Table 30: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.7e+08/ 1.5e+08	5.0e+08 /5.9e+08	3.1e+05 /7.4e+05	3.9e+04 /4.7e+04	4.0e+03 /1.2e+04	2.5e+03/ 2.3e+03	5.2e+05/ 4.2e+05
sumTree	F	4.6e+07 /5.7e+07	2.2e+08 /2.6e+08	6.8e+03 /3.7e+04	3.6e+02/ 3.0e+02	7.2e+02 /4.0e+03	3.2e+02/ 2.6e+02	8.7e+05/ 7.0e+05
sumTree TailRec	F	4.4e+07 /5.7e+07	1.9e+08 /2.2e+08	6.1e+03 /2.3e+04	3.5e+02 /3.5e+02	7.2e+02 /4.0e+03	3.2e+02/ 3.2e+02	8.7e+05/ 7.0e+05
Total		2.6e+08/2.6e+08	9.1e+08/1.1e+09	3.2e+05/8.0e+05	3.9e+04/4.8e+04	5.4e+03/2.0e+04	3.1e+03/2.9e+03	2.3e+06/1.8e+06

Table 31: Per-pass PAPI counters for `MonoTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.4e+08 /5.0e+08	7.1e+08 /1.2e+09	2.2e+05 /1.0e+06	3.4e+04 /5.4e+04	2.9e+03 /1.4e+04	1.6e+03 /3.3e+03	4.2e+05 /4.3e+05
sumTree	F	2.0e+08 /4.1e+08	4.3e+08 /6.3e+08	7.0e+04 /9.4e+05	3.3e+02 /3.4e+02	1.0e+03 /1.5e+04	3.0e+02/ 2.8e+02	3.8e+05 /4.0e+05
sumTree TailRec	F	2.0e+08 /4.1e+08	4.1e+08 /6.4e+08	6.5e+04 /9.3e+05	4.3e+02/ 2.6e+02	9.4e+02 /1.5e+04	3.9e+02/ 2.3e+02	3.7e+05 /4.0e+05
Total		6.5e+08/1.3e+09	1.6e+09/2.5e+09	3.6e+05/2.9e+06	3.5e+04/5.4e+04	4.8e+03/4.4e+04	2.3e+03/3.8e+03	1.2e+06/1.2e+06

Table 32: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
countLargeItems	F	3/5	40%	0.0133	0.0444	0.0109	0.0466	1.22 $\times$	3.33 $\times$	4.07 $\times$	0.95 $\times$
countMarked	F	3/5	40%	0.0134	0.0445	0.0146	0.0472	0.91 $\times$	3.33 $\times$	3.04 $\times$	0.94 $\times$
countSurvivors	F	4/5	20%	0.0142	0.0448	0.0128	0.0469	1.11 $\times$	3.15 $\times$	3.49 $\times$	0.96 $\times$
deadBytes	F	4/5	20%	0.0139	0.0447	0.0123	0.0471	1.13 $\times$	3.21 $\times$	3.64 $\times$	0.95 $\times$
liveBytes	F	4/5	20%	0.0137	0.0433	0.0123	0.0472	1.11 $\times$	3.15 $\times$	3.51 $\times$	0.92 $\times$
sumObjIds	F	3/5	40%	0.0133	0.0432	0.0103	0.0472	1.28 $\times$	3.26 $\times$	4.18 $\times$	0.91 $\times$
sweepUnmarked	M	5/5	0%	0.0747	0.0903	0.0814	0.1888	0.92 $\times$	1.21 $\times$	1.11 $\times$	0.48 $\times$
totalHeapSize	F	3/5	40%	0.0134	0.0650	0.0185	0.0504	0.73 $\times$	4.84 $\times$	3.51 $\times$	1.29 $\times$
touchHotObjects	M	5/5	0%	0.0729	0.0873	0.0829	0.1931	0.88 $\times$	1.20 $\times$	1.05 $\times$	0.45 $\times$
Total				0.2429	0.5074	0.2561	0.7146	0.95 $\times$	2.09 $\times$	1.98 $\times$	0.71 $\times$
Geomean								1.02 $\times$	2.72 $\times$	2.77 $\times$	0.83 $\times$

Table 33: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.7e+07/ 5.4e+07	2.6e+08 /2.8e+08	1.3e+06/ 4.7e+04	4.2e+02/ 2.9e+02	3.4e+03 /3.8e+03	4.0e+02/ 2.6e+02	2.9e+06/ 7.1e+05
countMarked	F	6.8e+07/ 5.6e+07	2.6e+08 /2.8e+08	1.3e+06/ 4.6e+04	3.8e+02/ 3.7e+02	3.4e+03 /3.9e+03	3.6e+02/ 3.3e+02	2.9e+06/ 6.8e+05
countSurvivors	F	7.2e+07/ 6.5e+07	3.1e+08 /3.8e+08	8.9e+05/ 6.0e+04	3.4e+02/ 2.7e+02	2.4e+03 /5.3e+03	3.2e+02/ 2.4e+02	2.8e+06/ 1.3e+06
deadBytes	F	7.1e+07/ 6.2e+07	2.7e+08 /3.5e+08	7.9e+05/ 5.1e+04	3.7e+02/ 3.5e+02	2.6e+03 /5.5e+03	3.5e+02/ 3.2e+02	2.7e+06/ 1.3e+06
liveBytes	F	7.0e+07/ 6.3e+07	2.7e+08 /3.5e+08	9.3e+05/ 5.1e+04	2.5e+02 /3.1e+02	2.6e+03 /5.6e+03	2.2e+02 /2.9e+02	2.8e+06/ 1.3e+06
sumObjIds	F	6.7e+07/ 5.2e+07	2.5e+08 /2.7e+08	1.2e+06/ 5.9e+04	3.2e+02/ 2.2e+02	3.1e+03 /4.3e+03	3.2e+02/ 1.8e+02	2.8e+06/ 6.9e+05
sweepUnmarked	M	1.9e+08 /2.4e+08	5.5e+08 /8.8e+08	1.1e+06/ 4.8e+05	8.9e+04 /1.1e+05	6.4e+03 /1.1e+04	3.3e+03 /7.6e+03	1.9e+06/ 9.9e+05
totalHeapSize	F	6.7e+07/ 5.1e+07	2.5e+08 /2.7e+08	1.3e+06/ 3.9e+05	7.4e+02/ 7.0e+02	2.9e+03 /9.0e+03	6.1e+02/ 5.5e+02	2.8e+06/ 5.9e+05
touchHotObjects	M	1.9e+08 /2.3e+08	5.7e+08 /9.0e+08	1.3e+06/ 4.7e+05	1.3e+05 /1.3e+05	5.3e+03 /1.1e+04	4.6e+03 /6.6e+03	1.8e+06/ 1.0e+06
Total		8.6e+08/8.7e+08	3.0e+09/4.0e+09	1.0e+07/1.7e+06	2.2e+05/2.4e+05	3.2e+04/5.9e+04	1.1e+04/1.6e+04	2.3e+07/8.5e+06

Table 34: Per-pass PAPI counters for `ObjectGraph` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.4e+05/ 4.7e+05	3.6e+02/ 2.3e+02	4.7e+03 /8.5e+03	3.3e+02/ 2.1e+02	1.8e+06/ 4.2e+05
countMarked	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.4e+05 /8.0e+05	1.7e+02 /1.9e+02	4.7e+03 /1.6e+04	1.6e+02 /1.6e+02	1.8e+06/ 4.1e+05
countSurvivors	F	2.3e+08 /2.4e+08	5.5e+08 /8.2e+08	1.7e+05 /3.7e+05	2.2e+02 /4.3e+02	1.8e+03 /7.1e+03	2.0e+02 /4.3e+02	1.8e+06/ 7.2e+05
deadBytes	F	2.3e+08 /2.4e+08	4.9e+08 /8.1e+08	3.6e+05 /3.9e+05	2.2e+02 /3.6e+02	3.6e+03 /7.2e+03	1.9e+02 /3.4e+02	1.8e+06/ 7.3e+05
liveBytes	F	2.2e+08 /2.4e+08	5.0e+08 /8.1e+08	1.4e+05 /3.8e+05	2.2e+02 /4.4e+02	1.8e+03 /7.2e+03	1.8e+02 /4.4e+02	1.8e+06/ 7.3e+05
sumObjIds	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.3e+05/ 4.7e+05	3.2e+02/ 2.0e+02	4.4e+03 /9.3e+03	3.1e+02/ 1.9e+02	1.8e+06/ 4.2e+05
sweepUnmarked	M	2.7e+08 /7.8e+08	7.6e+08 /2.1e+09	1.9e+05 /2.4e+06	7.6e+04 /1.4e+05	3.3e+03 /1.6e+04	3.2e+03 /6.1e+03	1.8e+06/ 9.3e+05
totalHeapSize	F	2.3e+08 /2.5e+08	4.4e+08 /7.8e+08	1.5e+06/ 1.4e+06	5.0e+02 /8.0e+02	2.2e+04 /2.6e+04	3.3e+02 /6.1e+02	1.5e+06/ 3.5e+05
touchHotObjects	M	2.7e+08 /8.0e+08	7.7e+08 /2.2e+09	2.6e+05 /1.8e+06	8.2e+04 /1.2e+05	2.9e+03 /2.2e+04	3.2e+03 /4.9e+03	1.7e+06/ 9.4e+05
Total		2.1e+09/3.3e+09	4.9e+09/9.9e+09	4.2e+06/8.5e+06	1.6e+05/2.6e+05	5.0e+04/1.2e+05	8.2e+03/1.3e+04	1.6e+07/5.7e+06

Table 35: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
addConstPW	M	8/8	0%	0.1272	0.1378	0.1412	0.4119	0.90 $\times$	1.08 $\times$	0.98 $\times$	0.33 $\times$
autorefineMaxLevel	F	4/8	50%	0.0200	0.0454	0.0190	0.2354	1.05 $\times$	2.27 $\times$	2.40 $\times$	0.19 $\times$
compressMass	F	3/8	62%	0.0190	0.0458	0.0115	0.2548	1.65 $\times$	2.41 $\times$	3.97 $\times$	0.18 $\times$
diffPW	M	8/8	0%	0.1253	0.1347	0.1332	0.4199	0.94 $\times$	1.07 $\times$	1.01 $\times$	0.32 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0208	0.0479	0.0185	0.2383	1.13 $\times$	2.30 $\times$	2.60 $\times$	0.20 $\times$
norm2Estimate	F	5/8	38%	0.0247	0.0650	0.0233	0.3689	1.06 $\times$	2.63 $\times$	2.79 $\times$	0.18 $\times$
pmapCutHistogram	F	4/8	50%	0.0316	0.0562	0.0124	0.2433	2.54 $\times$	1.78 $\times$	4.51 $\times$	0.23 $\times$
truncateTolViolations	F	3/8	62%	0.0192	0.0469	0.0115	0.2024	1.67 $\times$	2.44 $\times$	4.07 $\times$	0.23 $\times$
Total				0.3879	0.5797	0.3706	2.375	1.05 $\times$	1.49 $\times$	1.56 $\times$	0.24 $\times$
Geomean								1.29 $\times$	1.90 $\times$	2.44 $\times$	0.23 $\times$

Table 36: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.7e+08 /3.6e+08	5.7e+08 /1.2e+09	1.5e+06 /2.0e+06	2.6e+05 /2.6e+05	1.3e+04 /2.6e+04	6.6e+03 /2.6e+04	4.5e+06/ 1.8e+06
autorefineMaxLevel	F	1.0e+08/ 9.6e+07	2.5e+08 /3.7e+08	2.4e+06/ 1.9e+05	7.5e+02/ 1.5e+02	5.6e+03 /6.3e+03	7.4e+02/ 1.3e+02	6.1e+06/ 1.0e+06
compressMass	F	9.7e+07/ 5.9e+07	2.5e+08 /2.8e+08	2.1e+06/ 3.1e+04	6.9e+02/ 4.4e+02	2.0e+04/ 5.1e+03	7.0e+02/ 4.2e+02	6.3e+06/ 7.0e+05
diffPW	M	2.9e+08 /3.4e+08	5.9e+08 /1.3e+09	7.7e+05 /2.3e+06	1.5e+05 /2.0e+05	1.0e+04 /2.8e+04	9.7e+03 /2.7e+04	4.1e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.0e+08/ 9.4e+07	3.3e+08 /4.9e+08	2.4e+06/ 4.9e+05	6.2e+02/ 2.0e+02	1.4e+04/ 8.8e+03	6.1e+02/ 1.9e+02	6.3e+06/ 1.5e+06
norm2Estimate	F	1.2e+08/ 9.0e+07	3.4e+08 /4.4e+08	1.4e+06/ 1.1e+06	7.3e+02/ 5.2e+02	1.1e+04 /2.1e+04	6.5e+02/ 4.0e+02	6.2e+06/ 1.2e+06
pmapCutHistogram	F	1.6e+08/ 6.3e+07	2.8e+08 /3.7e+08	1.7e+06/ 5.8e+04	1.2e+03/ 3.5e+02	1.1e+05/ 7.1e+03	1.2e+03/ 3.3e+02	4.8e+06/ 1.3e+06
truncateTolViolations	F	9.8e+07/ 5.8e+07	2.5e+08 /2.9e+08	2.1e+06/ 4.7e+04	8.5e+02/ 3.6e+02	1.7e+04/ 4.8e+03	8.2e+02/ 3.4e+02	6.4e+06/ 7.1e+05
Total		1.2e+09/1.2e+09	2.8e+09/4.7e+09	1.4e+07/6.2e+06	4.2e+05/4.6e+05	2.0e+05/1.1e+05	2.1e+04/5.4e+04	4.5e+07/1.0e+07

Table 37: Per-pass PAPI counters for `PiecewiseFunctions` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
addConstPW	M	3.3e+08 /1.7e+09	7.8e+08 /3.3e+09	4.1e+05 /5.2e+06	1.3e+05 /3.4e+05	6.4e+03 /3.9e+04	4.7e+03 /4.4e+04	4.2e+06/ 1.9e+06
autorefineMaxLevel	F	2.3e+08 /1.2e+09	4.9e+08 /1.3e+09	2.9e+05 /2.1e+06	3.2e+02 /1.5e+03	1.5e+03 /4.1e+04	3.0e+02 /6.5e+02	4.3e+06/ 7.2e+05
compressMass	F	2.3e+08 /1.2e+09	4.7e+08 /1.3e+09	3.0e+05 /1.3e+06	2.8e+02 /1.0e+04	1.6e+03 /3.0e+04	2.5e+02 /6.2e+02	4.1e+06/ 4.2e+05
diffPW	M	3.5e+08 /1.8e+09	8.1e+08 /3.3e+09	3.9e+05 /5.3e+06	1.2e+05 /2.6e+05	5.6e+03 /4.9e+04	7.2e+03 /3.4e+04	4.1e+06/ 1.8e+06
lbDeuxLoadProxy	F	2.4e+08 /1.2e+09	5.3e+08 /1.3e+09	1.5e+05 /2.6e+06	2.8e+02 /5.8e+02	1.7e+03 /4.2e+04	2.6e+02 /5.2e+02	4.2e+06/ 1.0e+06
norm2Estimate	F	3.0e+08 /1.1e+09	5.1e+08 /1.3e+09	2.3e+06 /2.6e+06	5.4e+02 /2.4e+03	4.3e+04 /4.5e+04	4.3e+02 /1.5e+03	3.8e+06/ 1.0e+06
pmapCutHistogram	F	2.8e+08 /1.2e+09	5.0e+08 /1.3e+09	7.7e+05 /2.0e+06	7.7e+02 /2.0e+03	1.0e+04 /3.7e+04	7.3e+02/ 4.4e+02	4.0e+06/ 7.1e+05
truncateTolViolations	F	2.4e+08 /1.0e+09	4.4e+08 /1.3e+09	3.0e+05 /1.3e+06	4.9e+02 /1.8e+03	3.1e+03 /2.7e+04	4.3e+02 /5.8e+02	4.1e+06/ 4.3e+05
Total		2.2e+09/1.1e+10	4.5e+09/1.4e+10	4.9e+06/2.2e+07	2.6e+05/6.2e+05	7.3e+04/3.1e+05	1.4e+04/8.3e+04	3.3e+07/8.0e+06

Table 38: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add 1 tree	M	5/5	0%	0.0870	0.1135	0.0915	0.3738	0.95 $\times$	1.30$\times$	1.24$\times$	0.30 $\times$
sum tree	F	5/5	0%	0.0284	0.0596	0.0257	0.2737	1.10$\times$	2.10$\times$	2.32$\times$	0.22 $\times$
Total				0.1154	0.1731	0.1171	0.6474	0.99 $\times$	1.50$\times$	1.48$\times$	0.27 $\times$
Geomean								1.03 $\times$	1.66$\times$	1.70$\times$	0.26 $\times$

Table 39: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	3.8e+05 /1.1e+06	6.6e+04 /1.2e+05	3.2e+03 /1.3e+04	3.5e+03/ 2.1e+03	1.6e+06/ 1.1e+06
sum tree	F	1.4e+08/ 1.3e+08	4.6e+08 /5.4e+08	9.1e+04 /3.4e+05	1.0e+02 /1.1e+02	7.7e+02 /9.0e+03	8.1e+01 /9.4e+01	1.8e+06/ 1.4e+06
Total		4.1e+08/3.6e+08	1.2e+09/1.5e+09	4.7e+05/1.5e+06	6.6e+04/1.2e+05	3.9e+03/2.2e+04	3.6e+03/2.2e+03	3.3e+06/2.5e+06

Table 40: Per-pass PAPI counters for `TernaryTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.5e+08 /1.6e+09	9.5e+08 /2.4e+09	3.1e+06/ 2.3e+06	6.8e+04 /2.3e+05	5.2e+04/ 4.5e+04	4.6e+03/ 2.1e+03	1.2e+06/ 9.6e+05
sum tree	F	3.0e+08 /1.4e+09	5.8e+08 /1.0e+09	3.0e+06/ 2.3e+06	1.8e+02 /1.8e+03	4.9e+04/ 4.0e+04	1.6e+02 /1.1e+03	8.1e+05 /9.5e+05
Total		6.5e+08/3.0e+09	1.5e+09/3.4e+09	6.1e+06/4.6e+06	6.8e+04/2.3e+05	1.0e+05/8.5e+04	4.7e+03/3.2e+03	2.1e+06/1.9e+06

Table 41: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
autocompleteTopKProxy	F	5/10	50%	0.0160	0.0274	0.0086	0.1069	1.86$\times$	1.71$\times$	3.18$\times$	0.26 $\times$
countLazyNodes	F	4/10	60%	0.0140	0.0277	0.0066	0.1080	2.13$\times$	1.98$\times$	4.21$\times$	0.26 $\times$
countTerminals	F	3/10	70%	0.0134	0.0258	0.0101	0.1002	1.32$\times$	1.93$\times$	2.55$\times$	0.26 $\times$
decayTrieStats	M	10/10	0%	0.0775	0.0839	0.0854	0.2392	0.91 $\times$	1.08 $\times$	0.98 $\times$	0.35 $\times$
resetTraversalState	M	10/10	0%	0.0757	0.0845	0.0816	0.2361	0.93 $\times$	1.12$\times$	1.04 $\times$	0.36 $\times$
sumPrefixFreq	F	3/10	70%	0.0136	0.0384	0.0120	0.1072	1.13$\times$	2.83$\times$	3.19$\times$	0.36 $\times$
sumSubtreeHints	F	3/10	70%	0.0133	0.0261	0.0080	0.1045	1.65$\times$	1.97$\times$	3.25$\times$	0.25 $\times$
Total				0.2235	0.3139	0.2125	1.002	1.05 $\times$	1.40$\times$	1.48$\times$	0.31 $\times$
Geomean								1.35$\times$	1.72$\times$	2.32$\times$	0.29 $\times$

Table 42: Per-pass PAPI counters for `Trie` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	8.1e+07/ 3.9e+07	1.3e+08 /2.1e+08	1.5e+06/ 4.4e+05	1.6e+02/ 8.7e+01	5.1e+04/ 4.8e+03	1.4e+02/ 6.6e+01	3.8e+06/ 8.0e+05
countLazyNodes	F	7.1e+07/ 3.3e+07	1.3e+08 /1.6e+08	1.3e+06/ 2.6e+04	2.0e+02/ 1.6e+02	1.5e+04/ 3.4e+03	1.6e+02/ 1.5e+02	4.1e+06/ 6.6e+05
countTerminals	F	6.8e+07/ 3.1e+07	1.0e+08 /1.2e+08	1.3e+06/ 6.4e+05	3.2e+02/ 2.1e+02	2.3e+04/ 1.3e+04	3.0e+02/ 1.9e+02	4.0e+06/ 2.5e+05
decayTrieStats	M	1.7e+08 /2.1e+08	3.5e+08 /7.9e+08	7.3e+05 /1.3e+06	9.5e+04 /1.1e+05	6.3e+03 /2.0e+04	2.9e+03 /2.6e+04	3.0e+06/ 1.2e+06
resetTraversalState	M	1.6e+08 /2.0e+08	2.8e+08 /7.0e+08	1.0e+06/ 8.9e+05	9.0e+04 /1.1e+05	7.6e+03 /1.3e+04	3.4e+03 /1.7e+04	3.3e+06/ 9.1e+05
sumPrefixFreq	F	6.9e+07/ 3.1e+07	1.1e+08 /1.2e+08	1.3e+06/ 6.4e+05	6.1e+02/ 4.2e+02	2.7e+04/ 1.3e+04	5.2e+02/ 3.1e+02	4.0e+06/ 2.2e+05
sumSubtreeHints	F	6.7e+07/ 3.0e+07	1.1e+08 /1.2e+08	1.4e+06/ 2.6e+05	3.3e+02/ 1.7e+02	2.7e+04/ 6.6e+03	3.0e+02/ 1.4e+02	4.0e+06/ 3.1e+05
Total		6.9e+08/5.7e+08	1.2e+09/2.2e+09	8.5e+06/4.2e+06	1.9e+05/2.2e+05	1.6e+05/7.3e+04	7.7e+03/4.3e+04	2.6e+07/4.3e+06

Table 43: Per-pass PAPI counters for **Trie** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
autocompleteTopKProxy	F	1.4e+08 /5.4e+08	2.3e+08 /7.5e+08	1.1e+06 /1.5e+06	2.1e+02 /9.9e+02	3.1e+03 /2.7e+04	2.0e+02 /7.5e+02	2.8e+06/ 4.9e+05
countLazyNodes	F	1.4e+08 /5.5e+08	2.2e+08 /7.6e+08	2.2e+06/ 1.1e+06	1.4e+02 /7.3e+02	2.6e+03 /2.2e+04	1.3e+02 /4.0e+02	2.7e+06/ 3.5e+05
countTerminals	F	1.3e+08 /5.1e+08	2.0e+08 /7.3e+08	1.2e+06/ 6.4e+05	2.8e+02 /1.1e+03	1.7e+03 /1.6e+04	2.7e+02/ 2.6e+02	2.8e+06/ 2.1e+05
decayTrieStats	M	1.9e+08 /9.8e+08	4.5e+08 /2.1e+09	3.5e+05 /5.1e+06	9.1e+04 /3.9e+05	3.5e+03 /3.5e+04	4.6e+03 /3.8e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	2.0e+08 /9.6e+08	3.7e+08 /2.0e+09	3.1e+05 /3.2e+06	1.1e+05 /1.6e+05	5.0e+03 /3.0e+04	4.2e+03 /4.3e+04	2.9e+06/ 9.7e+05
sumPrefixFreq	F	1.2e+08 /5.3e+08	2.1e+08 /7.3e+08	1.5e+06/ 6.3e+05	5.1e+02 /1.4e+03	1.4e+03 /1.6e+04	3.8e+02 /4.1e+02	2.5e+06/ 2.1e+05
sumSubtreeHints	F	1.3e+08 /5.3e+08	2.1e+08 /7.3e+08	1.4e+06/ 6.1e+05	1.6e+02 /9.1e+02	1.4e+03 /1.6e+04	1.5e+02 /2.3e+02	2.8e+06/ 2.1e+05
Total		1.1e+09/4.6e+09	1.9e+09/7.8e+09	7.9e+06/1.3e+07	2.0e+05/5.6e+05	1.9e+04/1.6e+05	9.8e+03/8.3e+04	2.0e+07/3.7e+06